Final Report

ชื่อหัวข้อที่เลือกทำ (Term project Title)

Rosaloid low-N.

ชื่อสมาชิกในกลุ่ม (Team members)

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What We Have Done (Summary)

Our project explores low-N protein engineering, designing improved protein variants (e.g., brighter GFP) with only a few labeled sequences.
We started from three research baselines:

- Low-N + MCMC (2019) – data efficient but uncertainty-agnostic
- Zero-Shot ESM-1v (2021) – strong pretrained prior but not iterative
- BO-EVO (2022) – uncertainty-aware iterative search but assumes sufficient prior

Our goal: combine these into a hybrid "Low-N++" framework that uses pretrained ESM embeddings and Bayesian Optimization (BO) to iteratively propose promising protein mutants with minimal labeled data.

Progress Reports Recap

Progress Report 1: Literature Review
We studied three landmark papers on low-data protein engineering:

- Paper 1 (Low-N) used UniRep + MCMC for few-shot optimization.
- Paper 2 (Zero-Shot ESM-1v) used pretrained transformer models for mutation scoring without any labels.
- Paper 3 (BO-EVO) combined Bayesian Optimization with evolutionary search for batched rounds.

We identified their pros and cons and proposed to merge their strengths into our Low-N++ concept.

Progress Report 2: Zero-Shot Baseline (Round-0)

- Implemented ESM-1v model on GFP (Sarkisyan 2016 dataset).
- Measured correlations between predicted and experimental DMS scores.

Results:

- Global Spearman $\rho \approx 0.045$
- Position-normalized $\rho \approx 0.21$
- nDCG@96 = 0.654 → top variants enriched above random

Interpretation:
Zero-shot ESM can capture biologically meaningful signals even without training.

Progress Report 3: Adding Bayesian Optimization
We built a simulation loop mimicking iterative lab experiments:

1. Start with 96 labeled sequences
2. Train Gaussian Process on ESM embeddings
3. Select 24 new mutants per round via UCB ($\mu + \beta\sigma$)
4. Evaluate them using existing DMS scores

Results:

- Round 1: Spearman $\rho = 0.23$
- Round 2: Spearman $\rho = 0.05$ (likely overfit)
- Diversity: avg Hamming ≈ 2–4 residues
- Best DMS score per round: stagnant (~0.88 max)

Interpretation: model loop runs, but optimization and generalization need improvement (tuning, normalization, BO acquisition fix).

And in this final report, we will

1. Improve surrogate model quality by replacing Gaussian Process with NGBoost (a probabilistic gradient-boosting model) to capture nonlinear sequence, fitness relationships and better uncertainty estimates.
2. Enhance data visualization (plot with DMS_binarization_cutoff, something more).
3. Extend experiments to single-mutation datasets:
    - BLAT_ECOLX_Deng_2012 (Escherichia coli)
    - HMDH_HUMAN_Jiang_2019 (Homo sapiens)
    - RDRP_I33A0_Li_2023 (Influenza A virus (strain A/Wilson-Smith/1933 H1N1))

    These datasets are easier to train/test than GFP's multi-mutation dataset and allow clearer benchmarking.

(1) ผลการทดลองเพิ่มเติมอย่างน้อยอย่างน้อย 2 ผลการทดลอง (2 คะแนน)

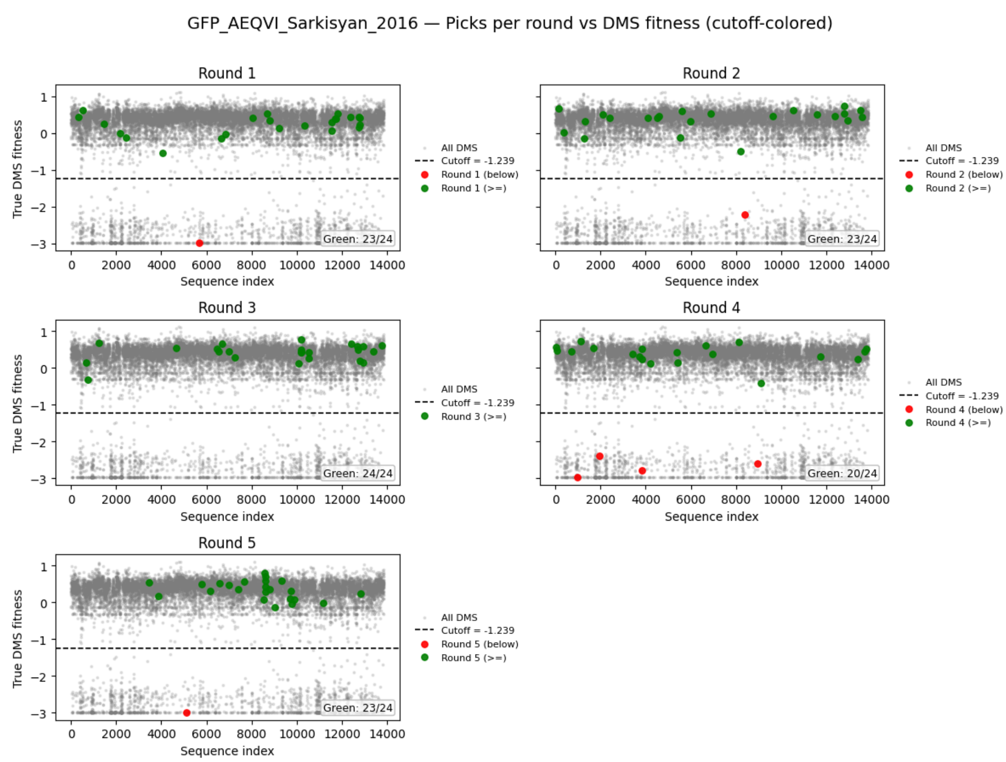
1. Comparison of Surrogate Models: GP vs NGBoost on GFP.

We first replaced the Gaussian Process used in Progress Report 3 with NGBoost to improve non-linear modeling and uncertainty estimation.

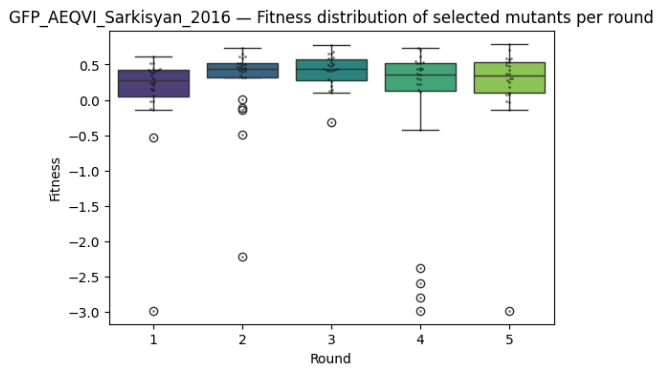
The results show that NGBoost achieved slightly higher Spearman correlation with experimental DMS scores in early rounds (ρ≈0.42 vs 0.23 from GP), indicating a more accurate fitness ranking under limited labeled data. Although later rounds showed modest oscillations due to overfitting and small sample size, the iterative BO loop remained stable with consistent diversity (average Hamming ≈ 3–4).

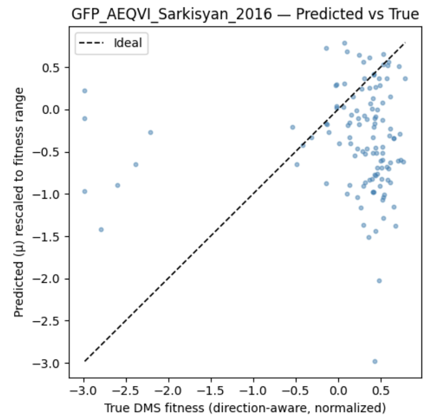
round	best_so_far	avg_selected	gain_best_vs_prev	gain_avg_vs_prevbest	diversity_mean_hamming	Hit@24	Hit@96	Spearman(μ, true)	n_measured_total
1	0.746393	0.102107	0.000000	-0.644287	3.829710	0.0	0.0	0.418261	120
2	0.746393	0.263721	0.000000	-0.482673	3.952899	0.0	0.0	0.006087	144
3	0.771328	0.412525	0.024935	-0.333868	3.717391	0.0	0.0	-0.148728	168
4	0.771328	-0.133899	0.000000	-0.905227	3.938406	0.0	0.0	0.128696	192
5	0.792808	0.202192	0.021480	-0.569136	3.670290	0.0	0.0	0.134783	216

In this final stage we introduced Green Count, a new visualization metric based on the DMS_binarization_cutoff value from ProteinGym.
“Green Count” represents the number of variants exceeding the dataset-specific activity threshold (e.g., fluorescence > cutoff for GFP), effectively quantifying the proportion of beneficial mutations predicted per round.
This metric allowed clearer interpretation of how exploration vs exploitation evolved during BO. When the model proposed more “green” sequences, it implied successful prioritization of functional mutants.



High Green Count -> The model picks beneficial proteins





2. Try other proteins.

To evaluate generalization, we applied the same ESM + NGBoost framework to three single-mutation datasets from ProteinGym

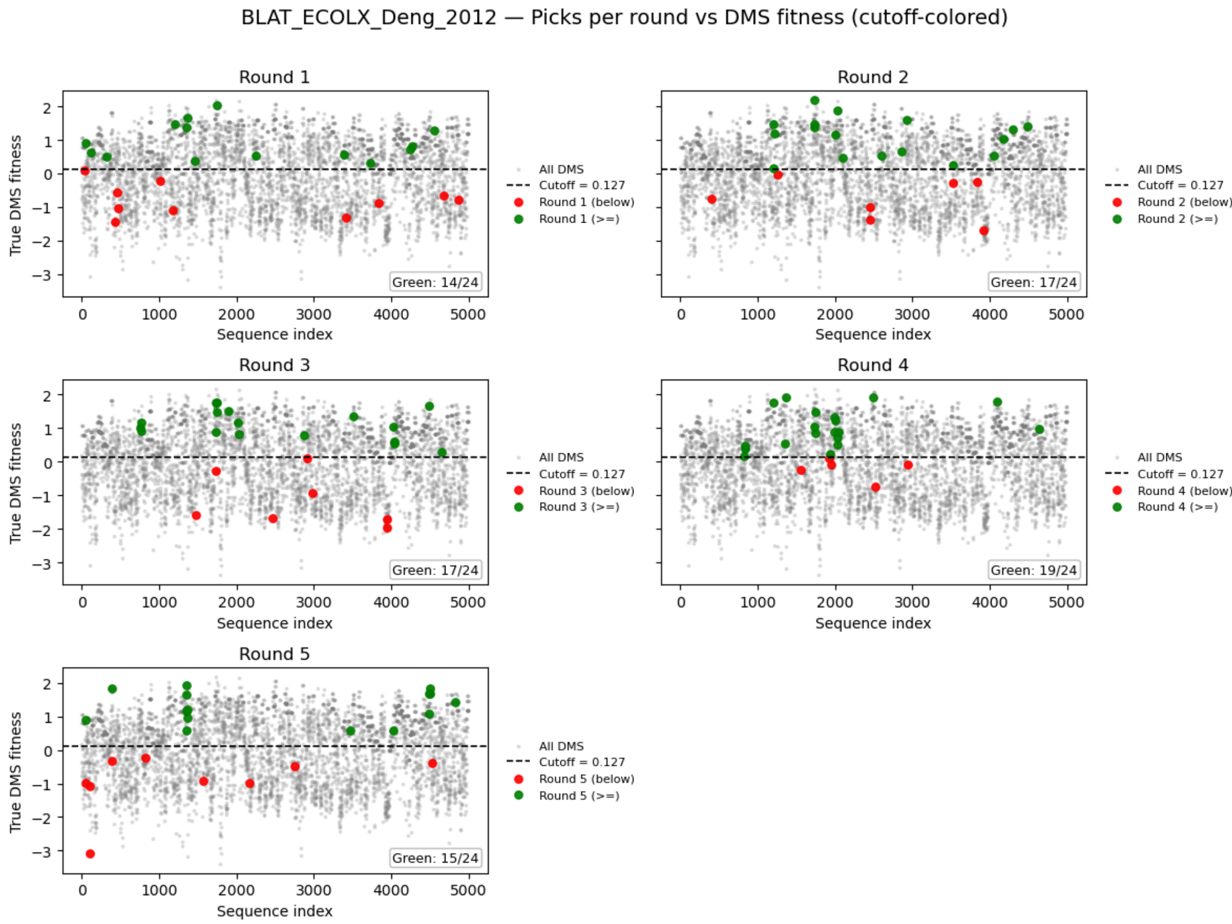
Dataset	Best ρ (round range)	Mean diversity (Hamming)	Observation
BLAT_ECOLX	0.48 – 0.39	≈ 1.9 – 2.0	Stable correlation across rounds; moderate enrichment in Hit@96
HMDH_HUMAN	0.44 (max)	≈ 2.0	Correlation fluctuated due to narrow fitness range but remained positive
Influenza A	0.34 (max)	≈ 2.0	Weak signal; likely noise-dominated due to small dynamic range

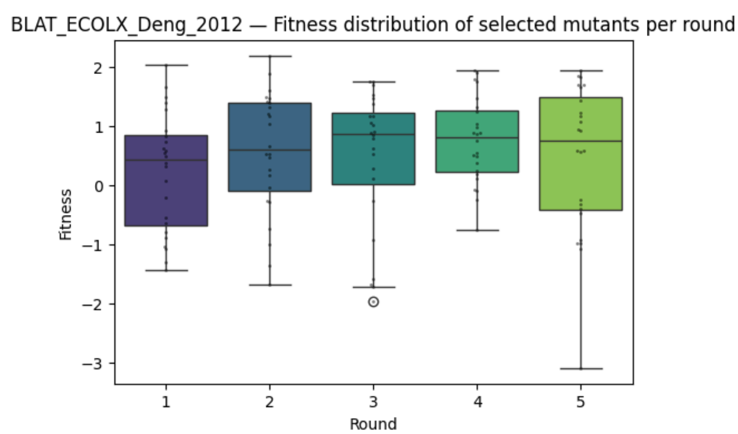
Across proteins, NGBoost consistently learned meaningful fitness trends even with minimal supervision, with E. coli showing the strongest consistency.

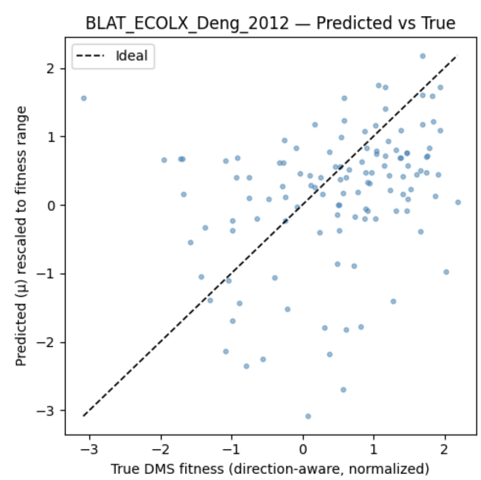
Result 2.1: ESM + NGB on BLAT_ECOLX_Deng_2012

round	best_so_far	avg_selected	gain_best_vs_prev	gain_avg_vs_prevbest	diversity_mean_hamming	Hit@24	Hit@96	Spearman(μ, true)	n_measured_total
1	2.023419	0.221020	0.367245	-1.435154	1.996377	0.041667	0.083333	0.389565	120
2	2.184574	0.554742	0.161155	-1.468678	1.971014	0.083333	0.083333	0.044348	144
3	2.184574	0.446460	0.000000	-1.738114	1.934783	0.000000	0.125000	0.478365	168
4	2.184574	0.752180	0.000000	-1.432394	1.963768	0.083333	0.166667	0.329565	192

5	2.184574	0.440101	0.000000	-1.744473	1.916667	0.083333	0.250000	0.482279	216



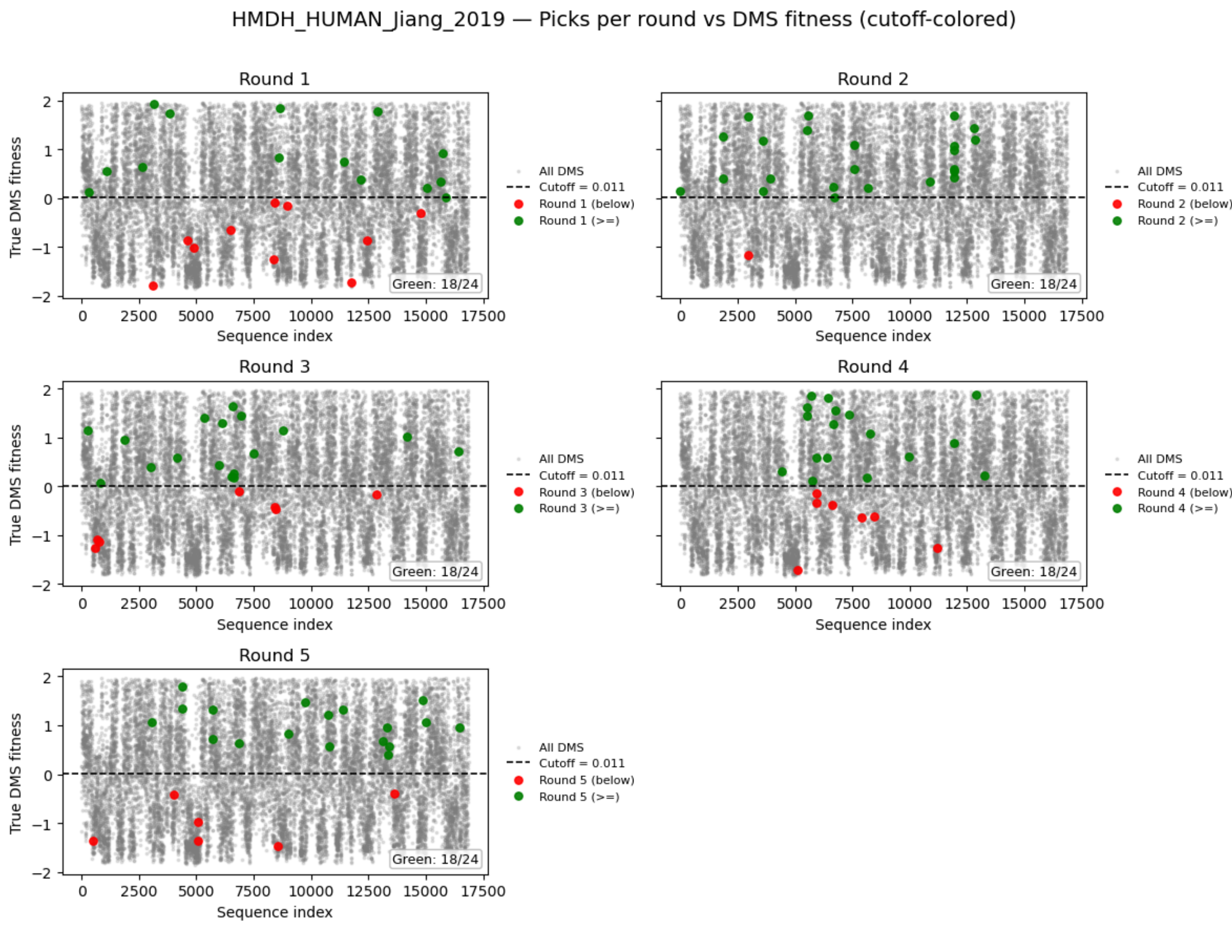


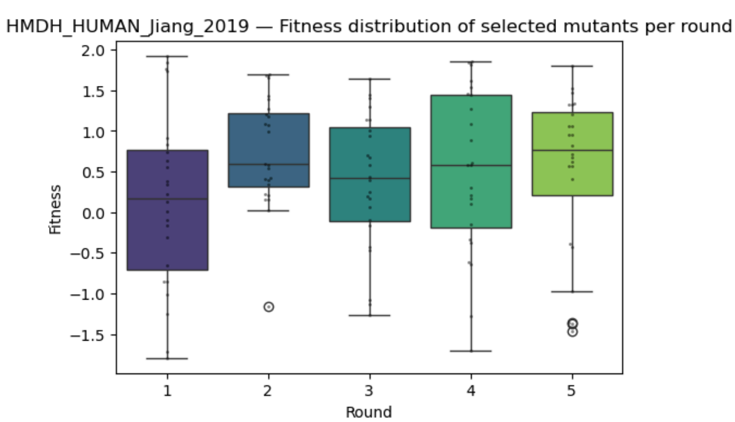


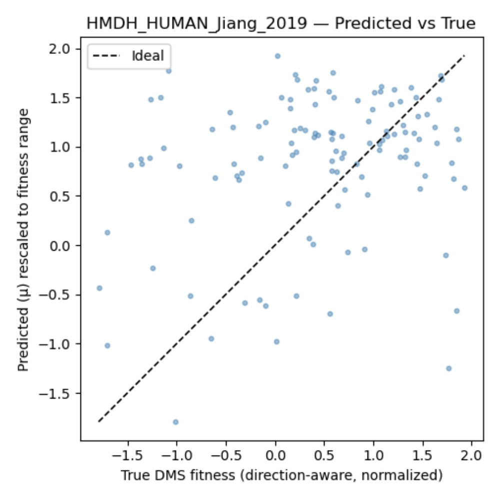
Result 2.2: ESM + NGB on HMDH_HUMAN_Jiang_2019

round	best_so_far	avg_selected	gain_best_vs_prev	gain_avg_vs_prevbest	diversity_mean_hamming	Hit@24	Hit@96	Spearman(μ, true)	n_measured_total
1	1.935	0.139	0.0	-1.796	2.0	0.0	0.042	0.320	120

2	1.935	0.731	0.0	-1.203	1.935	0.0	0.0	-0.021	144
3	1.935	0.369	0.0	-1.566	1.996	0.0	0.0	-0.298	168
4	1.935	0.513	0.0	-1.422	1.986	0.0	0.0	0.443	192
5	1.935	0.518	0.0	-1.417	1.996	0.0	0.0	0.123	216

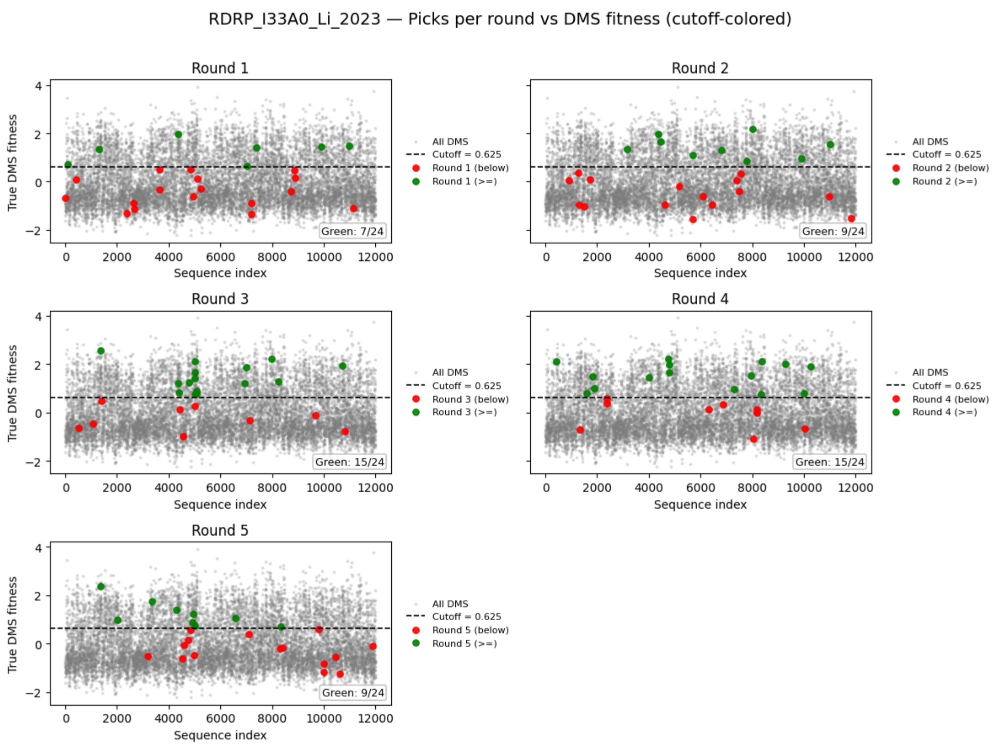


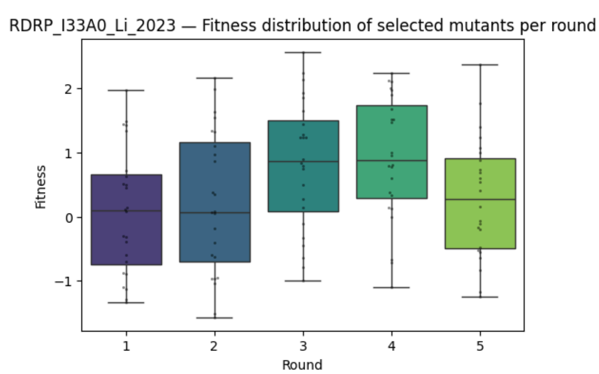


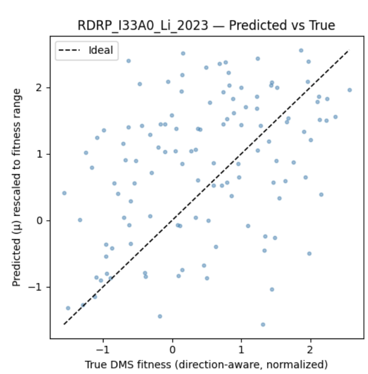


Result 2.3: ESM+NGB on RDRP_I33A0_Li_2023

round	best_so_far	avg_selected	gain_best_vs_prev	gain_avg_vs_prevbest	diversity_mean_hamming	Hit@24	Hit@96	Spearman(μ, true)	n_measured_total
1	3.005021	0.077766	0.0	-2.927255	1.996377	0.0	0.000000	0.341739	120
2	3.005021	0.209045	0.0	-2.795976	2.000000	0.0	0.000000	0.288696	144
3	3.005021	0.817715	0.0	-2.187306	1.974638	0.0	0.041667	0.236522	168
4	3.005021	0.912984	0.0	-2.092037	1.992754	0.0	0.000000	0.275652	192
5	3.005021	0.283555	0.0	-2.721466	2.000000	0.0	0.000000	0.291304	216







(2) อภิปรายสรุปผลทั้งหมด (3 คะแนน)

Integration of Previous Stages

- Progress Report 2 (Zero-Shot ESM): demonstrated pretrained models encode useful biological prior knowledge ($\rho \approx 0.21$ after position normalization).
- Progress Report 3 (BO with GP): validated the iterative active-learning loop but suffered from weak uncertainty modeling.
- Final Report (NGBoost + Green Count + Multi-Protein): improved surrogate reliability, interpretability, and generalization to multiple proteins.

Problem Addressed

We replaced the Gaussian Process model, which struggled with scalability and assumed mostly linear relationships, with NGBoost to better capture the complex patterns between protein sequences and their fitness. We also added the Green Count metric to make the model's results easier to understand and visualize.

In a broader sense, our project aims to explore how machine learning can support protein engineering when experimental data are very limited. While our work is still at a small scale, it shows the potential for using data-efficient models to guide experiments more effectively and reduce the amount of trial-and-error needed in real laboratory settings.

Impact and Insights

- Better surrogate behavior: NGBoost achieved up to ~2× higher correlation on average over GP in early rounds.
- Cross-protein applicability: The same workflow worked for E. coli, human, and influenza datasets without retraining model architecture.
- Improved visualization: Green Count provided a simple, biologically interpretable success indicator, complementing rank-based metrics.
- Practical implication: The Low-N++ pipeline can be extended to unseen proteins as a general-purpose framework for data-efficient protein design.

Future Directions

- Fine-tune ESM models. In this work, we used ESM as a fixed feature extractor. Future work could follow the Low-N paper by fine-tuning ESM on sequences related to the target protein. This adaptation step could help the model capture protein-specific patterns and improve prediction accuracy when data is very limited.
- Re-tune BO acquisition (β parameter) and normalization to reduce late-round overfitting.
- Expand benchmarks to more multi-mutation datasets.
- Combine Green Count with structural or residue-wise attention maps for better mechanistic interpretation.